

Now, defining $\lambda := p - 1$, $p_n(\cdot) := |F_n(\cdot)|$, $q_n(\cdot) := 2^p \cdot \max \left\{ r_{n-\frac{1}{2}}(\cdot), r_{n+\frac{1}{2}}(\cdot) \right\}$, for $n = 1, 2, \dots$ and $k_1 := 1$, $k_n := n - 1$, for $n = 2, 3, \dots$, one can easily see that Assumption (ii) of Theorem 17 is satisfied. Indeed, for $n = 1, 2, \dots$ it holds that

$$\begin{aligned} |f_n^p(t, \mathbf{u})| &\stackrel{(4.11)}{\leq} \left| r_{n+\frac{1}{2}}(t) \right| \cdot |u_{n+1} - u_n|^{p-1} + \left| r_{n-\frac{1}{2}}(t) \right| \cdot |u_n - u_{n-1}|^{p-1} + |F_n(t)| \\ &\stackrel{(4.14)}{\leq} 2^p \cdot \max \left\{ r_{n-\frac{1}{2}}(t), r_{n+\frac{1}{2}}(t) \right\} \cdot \max_{k=n-1, n, n+1} |u_k|^{p-1} + |F_n(t)|. \end{aligned}$$

The last three assumptions of Theorem 17 are obviously satisfied (due to Assumption (iii) and Assumption (iv) from Example 1). Thus, by Theorem 17, there exists a local solution of (4.13).

5. CONCLUDING REMARKS

Most of the results that can be found in the existing literature and concerning the question of existence of solutions for problems of type (3.1) require the assumption of $\mu(f(t, X)) \leq g(t) \cdot \mu(X)$, where $g(\cdot)$ is continuous and $\mu(\cdot)$ is either Hausdorff or Kuratowski measure of non-compactness. Our main result, Theorem 14, extends this assumption in two ways. Firstly, we require only $\mu(f(t, X)) \leq w_\alpha(t, \mu(X))$, with $w_\alpha(\cdot, \cdot)$ being an arbitrary Kamke function of order α (see Definition 8). Secondly, we allow $\mu(\cdot)$ to be an arbitrary measure of non-compactness satisfying some additional assumptions. Note that the above-mentioned $g(t) \cdot \mu(X)$ satisfies the definition of a Kamke function of order α (see Theorem 16 and Remark 16). Also, note that both the Hausdorff and Kuratowski measure of non-compactness satisfy given additional assumptions (the former case being discussed in Remark 10, the latter not mentioned in this paper, but one can easily find the necessary theory in [7]).

In Theorem 14, in contrast to a similar theorem for the integer order case in [4] (Theorem 13.3.1), we require the assumption of μ having the singleton property (see Definition 6). This is due to the fact that in (3.1), the initial condition is in a general point $u_0 \in E$, rather than in $u_0 = 0$. Moreover, as mentioned in Remark 9, this property does not follow from the definition of a (sublinear) measure of non-compactness in general, thus it indeed cannot be omitted.

Note that even though the theory was developed for fractional differential equations of order $0 < \alpha < 1$, the very same arguments can be used also in the case of $\alpha = 1$ (which coincides with the usual first derivative). However, note that for $\alpha = 1$, Theorem 14 is a known result (see the corresponding discussion in Section 1).

There are various options for further development of our main result, Theorem 14. Natural question is whether some assumptions of Theorem 14 can be weakened, at least for particular measures of non-compactness and/or particular Banach spaces. This question was discussed for the integer order case in [24], where it is shown that the mere continuity of f (instead of the uniform continuity) is sufficient in the case of a Hausdorff measure of non-compactness for some Banach spaces. Furthermore, the question of extension of local solutions to the whole interval $[a, a + \delta^*]$ can be addressed. Also, an interesting question is a generalisation of the IVP (3.1) in the context of calculus on time scales (even in the case of $\alpha = 1$). Finally, the IVP (3.1) can be also extended to fractional differential equations of arbitrary order $\alpha > 0$.

The application of Theorem 14 in the Banach space c_0 , namely Theorem 17, was inspired by [6], where a similar result is stated for the integer order case and $\lambda = 1$. However, we believe that in [6] (Theorem 3), the assumption of $\{p_j(\cdot)\}_{j=1}^\infty$ converging uniformly to